作业纸

班级： 姓名： 学号： 科目： 第 页

1) (a) 取 $f(x)=1, x, x^2$

$$\begin{cases} 2 = C_0 + C_1 + C_2 \\ 2 = C_1 + 2C_2 \\ \frac{8}{3} = C_1 + 4C_2 \end{cases} \Rightarrow \begin{cases} C_0 = \frac{1}{3} \\ C_1 = \frac{4}{3} \\ C_2 = \frac{1}{3} \end{cases}$$

$f(x)=x^3$

$\int_0^2 x^3\,dx = 4 = \frac{1}{3}\cdot 0 + \frac{4}{3}\cdot 1 + \frac{1}{3}\cdot 2^3$ 代数精度为3

$f(x)=x^4$

$\int_0^2 x^4\,dx = \frac{32}{5} \neq \frac{1}{3}\cdot 0 + \frac{4}{3}\cdot 1 + \frac{1}{3}\cdot 2^4$

(b) 取 $f(x)=1, x, x^2$

$$\begin{cases} 1 = C_0 + C_1 \\ \frac{1}{2} = C_1 x_1 \\ \frac{1}{3} = C_1 x_1^2 \end{cases} \Rightarrow \begin{cases} C_0 = \frac{1}{4} \\ C_1 = \frac{3}{4} \\ x_1 = \frac{2}{3} \end{cases}$$

代数精度为2

$f(x)=x^3$

$\int_0^1 x^3\,dx = \frac{1}{4} \neq \frac{1}{4}\cdot 0 + \frac{3}{4}\cdot\left(\frac{2}{3}\right)^3$

2) (a) 区间 $[0,\frac{1}{2}]$ $[\frac{1}{2},1]$ $h=\frac{1}{2}$

$I(f) = \frac{h}{6}\cdot\frac{1}{2}\cdot[f(0)+4f(\frac{1}{2})+f(1)+f(1)+4f(\frac{3}{2})+f(2)] = \frac{1179}{6800}$

(b) 区间 $[1,\frac{3}{2}]$ $[\frac{3}{2},2]$ $[2,\frac{5}{2}]$ $[\frac{5}{2},3]$ $h=\frac{1}{2}$

$I(f) = \frac{1}{6}\cdot\frac{1}{2}[f(1)+4f(\frac{5}{4})+2f(\frac{3}{2})+4f(\frac{7}{4})+2f(2)+4f(\frac{9}{4})+2f(\frac{5}{2})+4f(\frac{11}{4})+f(3)]$

$= 2.797427791$

3) (a) $[1,2]$ $[2,3]$ $h=1$

$I(f) = \frac{1}{6}[f(1)+4f(\frac{3}{2})+2f(2)+4f(\frac{5}{2})+f(3)] = 1.1$ $\quad 1\times10^{-2} > \varepsilon > 1\times10^{-3}$

(b) $[1,2]$ $[2,3]$ $f(x)=\frac{1}{x}$

$x = \frac{1}{2}t+\frac{3}{2}$ $\quad x=\frac{1}{2}t+\frac{5}{2}$

$t_0 = -\frac{\sqrt{3}}{3}$ $\quad t_{11} = \frac{\sqrt{3}}{3}$ $\quad A_0 = A_1 = 1$

$I(f) = \frac{1}{2}\left[\int_{-1}^{1} f(\frac{1}{2}t+\frac{3}{2})\,dt + \int_{-1}^{1} f(\frac{1}{2}t+\frac{5}{2})\,dt\right]$

$= \frac{1}{2}\left[1\cdot\frac{1}{\frac{1}{2}\cdot-\frac{\sqrt{3}}{3}+\frac{3}{2}} + 1\cdot\frac{1}{\frac{1}{2}\cdot\frac{\sqrt{3}}{3}+\frac{3}{2}} + 1\cdot\frac{1}{\frac{1}{2}\cdot\frac{\sqrt{3}}{3}+\frac{5}{2}} + 1\cdot\frac{1}{\frac{1}{2}\cdot-\frac{\sqrt{3}}{3}+\frac{5}{2}}\right] = 1.09771309$

$1\times10^{-3} > \varepsilon > 1\times10^{-4}$

WEIYANG COLLEGE TSINGHUA UNIVERSITY 清华大学未央书院

扫描全能王 创建

作业纸

班级： 姓名： 学号： 科目： 第 页

(4) 三点插值型

$f'(x_2) = -0.217$ $f'(x_3) = -0.195$ $f'(x_4) = -0.125$

$f'(x_1) \approx \frac{-3f(x_1)+4f(x_2)-f(x_3)}{2h} = -0.247$

$f'(x_5) \approx \frac{3f(x_5)-4f(x_4)+f(x_3)}{2h} = -0.179$

数值积分型

$f'(x_1) = -0.247$ $f'(x_5) = -0.179$

$$\begin{pmatrix} 4 & 1 & \\ 1 & 4 & 1 \\ & 1 & 4 \end{pmatrix}\begin{pmatrix} f'(x_2) \\ f'(x_3) \\ f'(x_4) \end{pmatrix} = \begin{pmatrix} -1.055 \\ -1.17 \\ -0.931 \end{pmatrix}$$

$f'(x_2) = -0.2156$

$f'(x_3) = -0.1924$

$f'(x_4) = -0.1846$

(5) $\int_0^1 \frac{1}{(x+1)\sqrt{x}}\,dx = \int_0^1 \frac{1}{\sqrt{x}}\,dx - \int_0^1 \frac{\sqrt{x}}{1+x}\,dx$

$= 2 - \frac{1-0}{6}[f(0)+4f(\frac{1}{2})+f(1)]$

$= 1.6024$

(6) $f(x)$ 在 $\frac{2}{3}$ 处 Taylor 展开：

$f(x) = f(\frac{2}{3}) + f'(\frac{2}{3})(x-\frac{2}{3}) + \frac{f''(\xi)}{2!}(x-\frac{2}{3})^2$

$\int_0^1 xf(x)\,dx = \int_0^1 *f(\frac{2}{3})\,x\,dx + \int_0^1 f'(\frac{2}{3})(x-\frac{2}{3})x\,dx + \int_0^1 \frac{f''(\xi)}{2!}(x-\frac{2}{3})^2 x\,dx$

$= \frac{1}{2}f(\frac{2}{3}) + 0 + \frac{1}{72}f''(\xi)$

(7) $\int_0^\infty e^{-x}x[L_m(x)-L_m'(x)]p_k(x)\,dx = \int_0^\infty e^{-x}xL_m(x)p_k(x)\,dx - \int_0^\infty e^{-x}xp_k(x)\,dL_m(x)$

$= \int_0^\infty e^{-x}xL_m(x)p_k(x)\,dx - [\int_0^\infty xp_ke^{-x}L_m(x)\,dx - \int_0^\infty e^{-x}p_k(x)L_m(x)\,dx$

$-\int_0^\infty e^{-x}xp_k'(x)L_m(x)\,dx]$

$= \int_0^\infty e^{-x}p_k(x)L_m(x)\,dx + \int_0^\infty e^{-x}xp_k'(x)L_m(x)\,dx$

$+$ 均为 0

$\therefore \int_0^\infty e^{-x}x(L_m(x)-L_m'(x))p_k(x)\,dx = 0$

对 $\forall p_{2n}(x)$，记其为 $p_{2n}(x) = xp_{2n-1}(x) + C$ $p_{2n}(0) = C$

$\int_0^\infty e^{-x}p_{2n}(x)\,dx = \int_0^\infty e^{-x}xp_{2n-1}(x)\,dx + C$

由上知，$L_m(x)-L_m'(x)$ 为以 $e^{-x}x$ 为权函数的正交多项式

x_k 为 L_n-L_n' 零点，$f[x, x_0, \cdots, x_n] \in P_{n-1}(x)$

$\therefore \int_0^\infty e^{-x}xP_{2n-1}(x)\,dx = \sum_{k=1}^n A_kP_{2n}(x_k)$

WEIYANG COLLEGE TSINGHUA UNIVERSITY 清华大学未央书院

扫描全能王 创建